Database Lab : PostgreSQL Connection String Diagnostic Automation

Duration : 30 Min.

Scenario

As a Database Engineer at NIIT, microservice deployment scripts frequently fail due to malformed PostgreSQL connection strings. You must author a diagnostic Python script `test_pg_dsn.py` that validates TCP port 5432 reachability and executes a database version query.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **PostgreSQL Service:** `localhost:5432` | DB: `postgres` | User: `postgres` | Password: `PostgresPassword123!`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Deliverable in `student_workspace/`: - **Diagnostic Script:** `test_pg_dsn.py`

Task Objectives

Author a Python diagnostic script `test_pg_dsn.py` inside `student_workspace/` that performs:

1. Socket Pre-Flight Check

- Implement a TCP socket reachability test confirming port 5432 on `localhost` is listening.

2. Live Database Handshake

- Connect to PostgreSQL (`localhost:5432`, db: `postgres`, user: `postgres`, password: `PostgresPassword123!`) using `psycopg2`.

3. Query Execution & Diagnostic Output

- Execute `SELECT current_database(), current_user, version();` and print the output string.

Verification

Run the verification script:

```
python run.py
```

The driver executes your script and validates socket testing and query output.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	test_pg_dsn.py Creation & TCP Socket Verification	4 Marks
TC2	PostgreSQL Live Driver Handshake on Port 5432	3 Marks
TC3	Query Execution (current_database & version) Output	3 Marks

Total Score: 10 Marks

Important Notes

- Ensure test_pg_dsn.py executes cleanly without errors.
- Use standard Python libraries (socket, psycopg2).